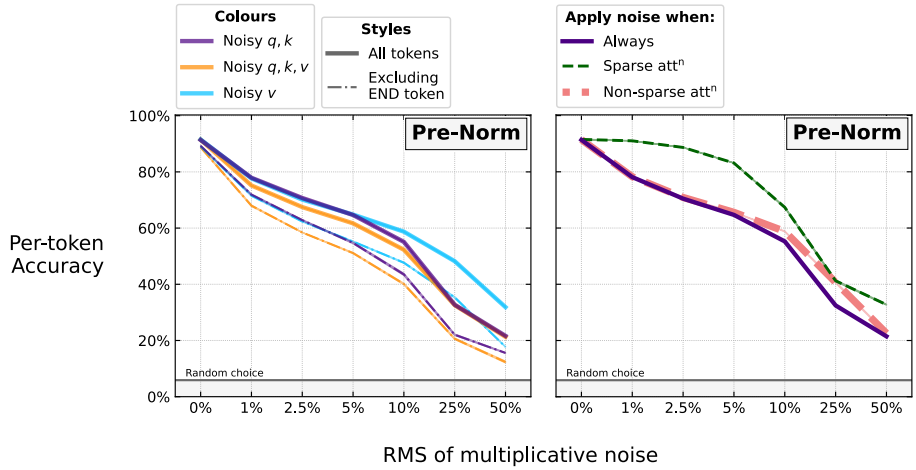

Figure 2: **Left:** evolution of per-token accuracy as we increase noise on the L_2-norms of $\{q, k_t, m_t\}$. A $\gtrsim 10\%$ drop in performance is observed when 1% noise is applied to all layers. **Right:** applying noise only to $\{q, k_t\}$, we see that non-sparse attention drives the drop at small noise, whereas the sparse case is stable. This is consistent with Theorems 5-6, but this interpretation is confounded by the relative importance of non-sparse distributions caused by frequency and depth-dependence.

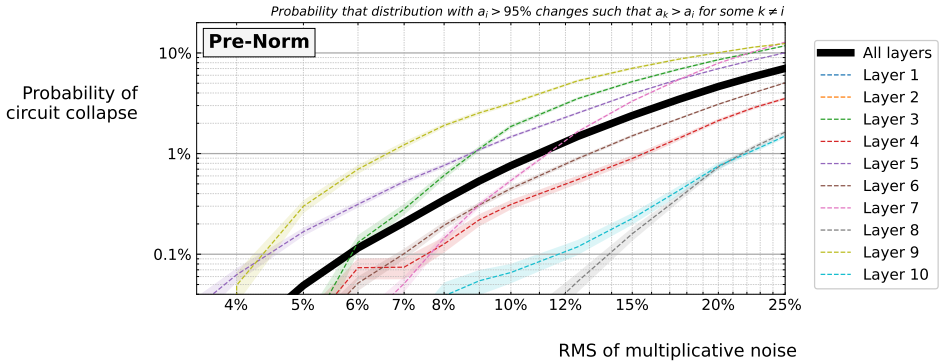

Figure 3: Probability of *circuit collapse* vs increasing noise. This observes the effect predicted in Section 6, and measures that 1% of sparse distributions collapse at a noise level of 11%.

8 Summary & Outlook

We have presented the idea that transformer `Pre-Norm` can cause interference between independent subspaces of the latent embeddings, a feature used by some real-world transformer circuits. Theoretically, we found this can only be avoided when using an embedding structure of *orthogonal spheres*. By contrast, the `QKV-Norm` architecture requires only linearly independent subspaces. We predict that sparse attention is stable with respect to interference, until a certain threshold of noise is reached, at which point it undergoes a phase transition called *circuit collapse*.

Empirically, we observe that the L_2-norms of attended embeddings are contained within a spread of $\pm 20\%$ for `Pre-Norm` (with 90% coverage), whilst `QKV-Norm` creates a wider spread. We simulate interference by introducing artificial noise onto the L_2-norms of $\{q, k_t, v_t\}$ in our trained `Pre-Norm` model, observing that 1% of sparse distributions collapse at a noise level of 11%. We observe that per-token accuracy degrades by $\mathcal{O}(10\%)$ when norms are simultaneously perturbed by noise of 1% in all layers, but is stable to %-scale noise in only sparse distributions. This may be attributed to either the predicted stability of sparse attention, or to a difference in the importance of sparse vs non-sparse heads induced by frequency and depth-dependence. More work is needed to disentangle these.

This work contributes a theoretical hypothesis of model behaviour, and empirically constrains the effect size without full model reverse-engineering. We have made predictions on representation structure, interference, and circuit collapse that practitioners may search for in their own models.

9 Limitations

We have not directly observed subspace independence or interference, and further work is required to establish their importance in real-world models. Experimentally, we simulate interference as being independent and similar in amplitude across heads and layers, however it is possible that it is correlated and depth-dependent. Whilst our stability experiments demonstrate that the model is more stable with respect to noise in sparse than non-sparse distributions, we have not shown whether this is due to the inherent stability of the attention distribution predicted by our theory, or the relative importance of sparse vs non-sparse distributions to the model. We show experimental results for a small model on a targeted task (with model variations in Appendix C); further work is needed to study the behaviour of larger models and different corpora.

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